The Hillingdon Hospitals
NHS Foundation Trust

Ultrasound in Obstetrics

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3.0	Ruma Dutta Consultant Obstetrician	Full review of guideline
4.0	Ruma Dutta Consultant Obstetrician, Terri Hodgkinson Lead Sonographer	Full review of guideline including • removing fetal RPD (moved to the joint paediatric renal anomaly guideline) • removing cervical length assessment (stand alone guideline) • removing indications for growth scans (NWL guideline introduced)

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1. Aim / Purpose

This guideline covers:

- Dating / Nuchal scan
- Anomaly scan
- Low lying placenta
- Growth scan measurements

2. Introduction

All women / birthing people referred for antenatal care at Hillingdon before 16 weeks are routinely offered two scans during pregnancy: a dating scan (which may include a nuchal translucency scan, depending on patient choice) between 11+2 to 14+1 weeks gestation and an anomaly scan between 18+0 to 20+6 weeks gestation. Women / birthing people referred after 16 weeks will be offered a combined dating and anomaly scan, with a follow up scan 2 weeks later to confirm the ultrasound EDD based on the 1st head circumference (HC) measurement.

All scans will be entered & reported on the Viewpoint software. A printed report should be filed in the ultrasound section of the maternal handheld notes. Blood results from the combined and quadruple tests will be entered on Viewpoint by administrative staff. The chance of a chromosomal disorder is NOT calculated using Viewpoint, an external laboratory system is used.

Informed consent for pregnancy ultrasound scans is required, for professional practice, and to ensure that women / birthing people are aware that although the vast majority of pregnancies are normal, the scan can bring bad news. With regards to safety, the sonographer should make every effort to minimise the length of time that the fetus is exposed to ultrasound. Publications are available for reference - BMUS Statement on the safe use, and potential hazards of diagnostic ultrasound (2002).

Subject to local infection control measures, the number of people accompanying a woman for her scan is limited to 1 adult plus their children, at the discretion of the sonographer.

3. The Guidance

3.1 Dating / Nuchal scan

3.1.1 Patients offered this scan

A dating scan is routinely offered to all women / birthing people booked before 16 weeks gestation at Hillingdon Hospital. This will include measuring the nuchal translucency (NT) as part of the combined screening test (CST) for trisomies 13/18/21, if the patient has given written consent for this test.

3.1.2 Booking

Booking of dating scans on ICE by booking midwife and an appointment is sent out by the ultrasound team and a screening blood test will be done by the antenatal clinic office once the referral has been received from the GP/midwife/patient. The appointments will be made at 12 weeks gestation by the stated last menstrual period (LMP) providing the referral has

been received by 9+0 weeks. Referrals received later than 9+0 weeks may result in the scan being booked later than 12 weeks. Referrals received after 12+0 weeks will be unlikely to have the scan booked in time for a combined screening test. These patients should be offered serum screening once they have had a dating scan. Referrals received after 16 weeks will be sent an anomaly scan appointment rather than a dating scan.

3.1.3 Number of scans offered

Women / birthing people are offered one dating scan. Women / birthing people who attend between 8+0 and 11+1 weeks will have a repeat scan only if they wish for combined screening for trisomies 13/18/21. If a nuchal measurement is not possible between 11+2 and 14+1 weeks, due to fetal position and/or patient size, then the sonographer will explain this to the patient and offer the quadruple test after 14+2 weeks. Repeat nuchal scans will not normally be offered.

3.1.4 Patient information & Consent

Information about screening for trisomies (either a link to National Screening Committee (NSC) website or a booklet “Screening tests for you and your baby”) is included in the initial information pack sent to the patient with their first appointment letter.

Discussion about combined testing must take place with the booking midwife and a consent form signed if the woman wishes to have the test. The form is filed in the maternity hand-held notes. It must be done before the patient attends for her ultrasound. In certain circumstances this may not have been possible. When the patient arrives, the sonographer must discuss the purpose and limitations of the scan and ensure informed consent is obtained, if required, including a signed form.

a) If the woman has already decided to have screening and is aware of the choices, then the sonographer will check the written informed consent form is available and offer the combined test. This will be documented on the Viewpoint report.

b) In cases where the woman has chosen not to have screening, the patient’s wishes must be noted on Viewpoint and in the handheld maternity notes.

Sonographers should take this opportunity to inform women / birthing people about their anomaly scan. They should ask them to read the relevant section in the screening booklet and request that the woman makes the appointment at the desk before leaving the department.

3.1.5 Dating Scan Technique

This is more commonly a transabdominal scan, some patients may need a transvaginal scan. Depending on gestation and maternal size, a basic anatomical survey should be carried out as detailed on Viewpoint. However if all categories cannot be completed, a repeat scan is not indicated, since the patient will routinely have an anomaly scan. The sonographer will measure the crown-rump length CRL in order to date the pregnancy.

If CRL <45.0mm – too early for screening. Arrange a repeat scan for approximately 12 weeks for NT measurement if the woman wishes to have the test.

If CRL 45.0mm-84.0mm, measure the NT – this is the appropriate window for the combined test. Complete the CST blood form. If NT is not possible, then offer the quadruple test after 14+2 weeks. Rescan is not necessary.

If CRL >84.0mm, this is too late for the combined screening test. Book the woman for a quadruple test. The HC needs to be measured for both dating and for the quadruple test. Complete the quadruple test blood form.

3.1.6 Calculation of estimated date of delivery (EDD)

The EDD should be calculated from the ultrasound measurement obtained at the dating scan or an early pregnancy scan above 8+0 weeks gestation using Viewpoint and entered on the antenatal clinic section of the maternal handheld notes. Any other EDD (e.g. from last menstrual period [LMP]) should be deleted. If the woman is already more than 23+0 weeks pregnant, ultrasound is less accurate for gestation calculation. Use the HC measurement on Viewpoint to calculate the EDD and book a repeat scan in 2 weeks to confirm forward growth.

If the woman has had a dating scan at another UK trust above 8+0 weeks, and has a written report for confirmation, the EDD will be adjusted accordingly.

3.1.7 Abnormal results

Fetal demise

The woman will be transferred to the Early Pregnancy Assessment Unit to be seen by the nurse or doctor.

Fetal abnormality

If an abnormality is detected, the guideline “Referral when a fetal abnormality is detected” should be followed. The first step will be to inform the lead sonographer or deputy. The patient will be referred to the specialist midwife and further action taken, depending on the anomaly.

3.1.8 Multiple pregnancy

Please refer to the guideline “Multiple Pregnancy”. At the dating scan, one image must be taken and stored which demonstrates both fetuses on the same image. The CRL of the larger fetus must be used to estimate gestational age, in order to avoid the risk of the smaller fetus having early growth pathology.

Chorionicity must be determined by 14 weeks and entered on Viewpoint. If chorionicity is not clear, a transvaginal scan can be considered. If not appropriate or uncertainty remains, please refer for a fetal medicine scan appointment with one of the consultants for obstetric ultrasound. When the chorionicity is determined, the sonographer should ensure that appropriate scan follow-up appointments are made. They should inform the specialist midwives if a pregnancy is found to be monochorionic diamniotic or monoamniotic (MCDA or MCMA).

With regards to the combined screening testing (CST) for trisomy 21 in a multiple pregnancy, after checking informed consent, nuchal translucency (NT) measurements are taken.

- If the scan shows a fetal pole with no heartbeat with one remaining viable fetus, then CST is not valid and the NT alone is used for screening. NT measurement risk must not be calculated by Viewpoint. The data is sent to an external laboratory by the specialist midwife.
- If there is an empty gestational sac and one or two remaining viable fetuses, then CST can be done.

The quadruple screen is available for twin pregnancies for Downs syndrome only. On diagnosis of a multiple pregnancy, the woman should be offered referral to the specialist

midwife for further counselling regarding the issues of invasive testing if the result comes back showing a high risk, if time allows.

3.2 Anomaly scan

Normally one scan is offered. Repeat scans may be necessary due to fetal position or limitations imposed by maternal obesity, fibroids and other conditions.

3.2.1 Indications for anomaly scan

This scan is routinely offered to all women / birthing people registered for maternity care at Hillingdon Hospital, between 18+0 and 20+6 weeks, and should be completed by 23+0 (FASP December 2021). Late bookers can have a limited anomaly scan, but the limitations of a late anomaly scan should be discussed and documented in the Viewpoint report. Repeat anomaly scans are not indicated if the woman has already been scanned in another NHS unit, privately or abroad, provided she has a copy of the report.

Women / birthing people with a high chance of a fetal anomaly may need an earlier targeted anomaly scan. The timing of this will vary with the suspected/possible anomaly and is booked on the fetal medicine list. It may also be appropriate to do additional anomaly scans at a later gestation. This should be discussed with the specialist midwife and/or one of the consultants for obstetric ultrasound.

3.2.2 Booking anomaly scans and late bookers

The scan should be booked by the ultrasound reception staff once the dating scan has established the EDD. In cases where the booking referral arrives at >16 weeks by LMP, a provisional anomaly scan appointment will be sent to the patient by the ANC office rather than a dating scan.

If the woman attends for her first scan outside the FASP anomaly scan guidelines i.e. later than 23+0 weeks, fetal anatomy should be checked as best as fetal position and gestation allow. The scan report should document that the woman was a late booker and she should be booked for repeat biometry after 2 weeks, as in 3.1.6.

3.2.3 Patient information and Consent

The informed consent process starts at booking with a discussion with the midwife, and signing a consent form. This should be checked prior to the scan by the sonographer. This form is filed in the maternity handheld notes. Before the scan is started, the sonographer should note whether or not the patient opted for screening for trisomies.

3.2.4 Recording the scan

The FASP guidelines 2021 advise that images of 6 specific anatomical sections should be archived. These are:

- Head circumference (HC) measurement and the atrium of the lateral ventricle.
- Suboccipitobregmatic view demonstrating measurement of the transcerebellar diameter
- Coronal view of lips with nasal tip
- Abdominal circumference (AC) measurement
- Femur length (FL) measurement
- Sagittal (preferred) or coronal view of the spine including the sacrum.

At Hillingdon Hospital, further images of the fetal anatomy should be recorded as shown in Appendix 1. All scans should be recorded on the Picture Archiving and Communication system (PACS) and entered on Viewpoint. Where appropriate, the structures should be assessed in sagittal, coronal and transverse planes.

3.2.5 Performing the scan

The anomaly scan includes a survey of the uterus, ovaries and adnexae. If there is a large mass or one which looks suspicious of a dermoid or a malignancy, the woman will be referred to a consultant-led antenatal clinic within a week.

Fetal measurements which must be measured and recorded are shown above in 3.2.4, HC, AC and FL. In addition, cerebellar diameter (TCD) and ventricular index (anterior and posterior) should be measured and recorded. All these are performed using standard techniques.

If the estimated fetal weight (EFW) at the anomaly scan is <10th centile, the woman must be referred to the next HRAN list and the FGR guideline must be followed.

If the EFW is >90th centile, book the woman a consultant-led antenatal clinic appointment at 28 weeks.

Placental site is reported as 'low' in the second trimester if it is within 2 cm of the internal os or covers it – see section 3.3 and Trust guideline "Placenta praevia incorporating accreta, increta and percreta: diagnosis and management". In cases of a previous caesarean section (CS) or previous myomectomy & an anterior low placenta, placenta accreta should be excluded. A suspicious or confirmed accreta must be discussed with the lead sonographer or deputy, and/or one of the consultants for obstetric ultrasound the same day. The woman may require additional scans or MRI later in pregnancy and so she needs a consultant review within 3 working days.

3.2.6 If the anomaly scan cannot be completed

If biometry and structural survey cannot be completed due to **fetal position** duing the allotted 30min timeframe, the woman will be asked to mobilise and then a further attempt made. If this is unsuccessful then a repeat appointment should be made as soon as possible.

If normal anatomy is not seen due to **maternal obesity**, a repeat scan should be booked by 22+6 weeks. If normal anatomy cannot be seen at the second scan the patient should be informed as to why it is not possible to complete the scan and this should be recorded clearly in the notes. A further repeat scan is not required.

However in some cases, such as a high risk woman with diabetes, when the anomaly scan remains incomplete, the case should be discussed with one of the specialist midwives and/or one of the consultants for obstetric ultrasound.

3.2.7 Abnormalities

If an abnormality is detected, the guideline "Referral when a fetal abnormality is detected" should be followed. The first step will be to inform the lead sonographer or deputy. In cases where a definite anomaly is seen, the sonographer should inform the parents of the findings. The woman is referred to the specialist midwife within 1 working day and further action is taken, depending on the anomaly – in most cases, the woman should be seen within 3 working days on a fetal medicine list with a consultant obstetrician and/or 5 working days at

a tertiary referral centre (FASP-S08). She should be given contact information for the specialist midwife, if the midwife is not immediately available to speak to her.

The scan must be recorded on Viewpoint as “Abnormal scan”. 3 copies of the report should be made:

- woman’s hand-held notes
- specialist midwife
- ultrasound anomalies folder.

In cases where an anomaly is suspected but not confirmed, the sonographer should seek the advice of the lead sonographer or deputy, or if they are not available, a more senior sonographer. If circumstances arise where there is no senior sonographer available and the findings are unsure then the scan should be reported as incomplete and rebooked at a time when more experienced staff are present (within 2 working days).

3.2.8 EDD adjustment

If the patient has not had a dating scan, she should be given an EDD using the HC measurement at her anomaly scan. A repeat scan for growth must be booked in two weeks. If growth is normal at the 2nd scan, in line with the previous scan, then the earlier EDD can be used, telling the patient that in most cases the baby will be born close to this date.

3.2.9 Multiple pregnancy

If chorionicity is not known prior to the anomaly scan, it may be possible to check for discordant fetal gender or completely separate placental masses. If chorionicity cannot be confidently confirmed, then the woman should be managed as having monochorionic twins ie she should be given an appointment on the fetal medicine scan list to see a consultant obstetrician after 2 weeks.

3.2.10 Intrauterine death

Inform the on call registrar or consultant to see the patient regarding further management. Find a room where the parents can sit in privacy. Refer to the guideline “Intrauterine death, Neonatal death and late termination of pregnancy: medical management”.

3.2.11 Fetal Sexing

The fetal gender is not routinely checked or recorded except in rare cases, for example where the gender is needed to determine the risk of fetal genetic disorders. At other times, parental request to inform them and/or write down the fetal gender is at the discretion of the sonographer and their wishes should be respected.

3.3 Low lying placenta

See section 3.2.5 above and refer to the guideline “Placenta praevia incorporating accreta, increta and percreta: diagnosis and management”.

3.3.1 Detected at the anomaly scan

The placenta is reported as low in the Viewpoint report and the woman booked a further scan at 32 weeks. The woman should be given the ‘Low lying placenta’ information sheet.

3.3.2 At 32-36 weeks

1. In cases of **asymptomatic women / birthing people with a low placenta,** imaging should be performed at around 32 weeks of gestation to clarify the diagnosis, including a transvaginal scan. If the placenta is still low at 32 weeks (within 2cm of the internal os), rebook for a 36 week scan and arrange a consultant-led antenatal clinic appointment on that same day in case delivery planning is needed. If the placenta is no longer low at 32 weeks

(more than 2cm from the internal os), the woman can return to normal midwifery or consultant-led care.

2. In cases of **asymptomatic women / birthing people with a low-lying placenta and a previous caesarean section or myomectomy**, there is a risk of placenta accreta – see section 3.2.5. If suspicious or confirmed accreta at 32 weeks, the case must be discussed with the lead sonographer or deputy, and/or one of the consultants for obstetric ultrasound the same day and seen by a consultant within 3 working days.

Greyscale antenatal sonographic imaging can be complemented by both colour Doppler sonographic imaging and magnetic resonance imaging (MRI) in equivocal cases. MRI can add additional information in terms of degree on placental invasion, particularly to adjacent organs, and therefore may assist surgical planning. The decision for MRI should be on a case by case basis with discussion between the consultant obstetrician and consultant radiologist.

3. If the patient is **admitted with an antepartum haemorrhage**, where clinically appropriate, a repeat ultrasound to confirm placental location may be considered. In the case of recurrent antepartum haemorrhage, fetal growth surveillance should be undertaken.

4. There is a place for TVS if the fetal head is engaged **prior to an otherwise planned caesarean section for placenta praevia**. Ultrasound can elucidate where the fetal head is relative to the leading edge of the placenta.

If vasa praevia is suspected or if the patient has had any PV bleeding then the woman should see the on call obstetric registrar.

3.4 Growth scans

3.4.1 Measurements taken at growth scan

Viewpoint must be used to document the following measurements:

- Head circumference
- Abdominal circumference
- Femur length
- Liquor volume –a deepest vertical pool measurement should be documented
- Umbilical artery dopplers including comment on presence, absence or reversal of end diastolic flow (EDF) - with an umbilical artery PI >95th centile and EDF present, or absent EDF or reversed EDF, the woman must be referred to day care unit or to the on call registrar the same day – see FGR guideline.

Middle cerebral artery dopplers should be performed if the umbilical artery doppler is close to or above the 95th centile, or if the umbilical artery Doppler shows absent or reversed EDF. In these situations, the woman must be referred to day care unit or to the on call registrar after the scan on the same day.

3.4.2 Indications for growth scans

The guideline “Identifying and Managing Fetal Growth Restriction (FGR)”, based on the NWL regional guideline, gives the criteria for risk assessment for all women / birthing people at the start of pregnancy and therefore who warrants scans for fetal growth +/- uterine artery Doppler scans.

Referral for a growth scan when the symphysio-fundal height (SFH) is small for dates (SFD) on the SFH chart is also covered in the FGR guideline.

3.4.3 Head Circumference

If the HC is below 5th centile, the woman should be referred to the specialist midwife and then a repeat scan booked on a fetal medicine scan list. The team must check the results for chromosomal screening tests, the anomaly scan images and offer a TORCH screen (also a ZIKA test if the travel history is relevant). Follow up and timing of repeat scans will depend on the individual case.

3.4.4 Short Femur

The femur length (FL) should routinely be measured as part of an anomaly scan and also at the growth scan. If the measurement is on the 5th centile, then the sonographer should book a repeat growth scan after 2 weeks, provided the anomaly scan is otherwise normal and the combined screening test (if done) gave a low chance result for the 3 trisomies.

If the FL is below the 5th centile, refer to the specialist midwife and to the next fetal medicine scan list. At this scan, the past obstetric and family history should be reviewed. The results of screening for chromosomal anomalies should be documented. The other long bones should be measured and all measurements documented on Viewpoint. Also check the following:

- Skull shape
- Spine
- Chest size and shape
- Echogenicity of bones
- Shape of bones
- Presence or absence of fractures
- Other anomalies
- Liquor volume
- Uterine artery doppler

Counselling should be offered by the specialist midwife and/or one of the consultants for obstetric ultrasound. This includes possible diagnoses:

- Constitutional
- FGR
- Skeletal dysplasia
- Genetic syndromes
- Chromosomal anomalies

Repeat scan should be offered depending on gestation and the overall findings.

4. Associated Guidelines, Policies and Documents

Multiple Pregnancy
Placenta Praevia Incorporating Accreta/Increta and Percreta: Diagnosis and Management
Referral when a Fetal Abnormality is Detected
Referral for Fetal Echocardiogram
Down's Syndrome, trisomy 13 and 18 – Screening and Diagnostic Testing
Intrauterine Death, Neonatal Death and Late Termination of Pregnancy: Medical Management
Fetal Renal and Bladder Anomalies – Antenatal Ultrasound and Neonatal Management
Identifying and Managing Fetal Growth Restriction (based on NWL guidelines)
Day Assessment Unit SOP

5. Training needs identified

The scanning will only be done by health professionals with a recognised postgraduate qualification in ultrasound. Staff in training will be under the supervision of a qualified sonographer, who is responsible for the accuracy of the scan. Medical staff with appropriate training & experience are able to perform scans, this should be agreed by their educational supervisor and their name added to the Viewpoint system by the lead sonographer or deputy so that they can record and report their scans.

Continuing professional development needs to be maintained as per HCPC requirements. Fetal Medicine Foundation registration which covers nuchal translucency measurements must be kept up to date by submitting yearly images. Annual Hillingdon Hospital trust appraisal takes place in the form of PDR.

6. Implementation and Monitoring

All guidelines, protocols and procedures are available on the intranet. All staff have a responsibility to check the Trust's intranet and ensure that they are referring to the most recent issue of the document. The weekly general information bulletin includes notification of publication of new/updated documents.

This clinical guideline, protocol or procedure will be monitored using local and National guidance and via the Trust Incident and Risk Management Policies. If risks/issues are identified further specific monitoring will be considered, for example, clinical audit or risk assessment

7. References

UK National Screening Committee guidelines – January 2016
NHS Fetal Anomaly Screening Programme: Consent Standards and Guidance – December 2021
RCOG Placenta Praevia and Placenta Accreta: Diagnosis and Management (Green-top Guideline No. 27a) – September 2018

8. Appendices

Appendix 1 – Fetal Anatomy Checklist

Appendix 2 – FASP 11 auditable conditions

Appendix 1

Fetal Anatomy Checklist at anomaly scan

Anatomy	**Scan findings**	**Images to record at THH**
Heart beat	The normal heart rate is 110-160 bpm. Sustained bradycardia of less than 100 bpm or sustained tachycardia of more than 180 bpm must be reported to the specialist midwife and/or a consultant while the patient is still in the department and referred to a tertiary centre within 24-48 hours	
Fetal movements	Normal movements should be seen.	No images
Presentation	Not normally recorded in early 2nd trimester	No images
Amniotic fluid	Deepest pool if fluid is NOT normal	Measurement of deepest vertical pool
Cord	Normal 3 vessels and abdominal insertion	Abdominal wall and cord insertion demonstrating Colour Doppler image of 2 umbilical arteries around fetal bladder or transverse view of the umbilical cord
Head	Check oval shape of skull	Head circumference at the level of Va (anterior ventricular horn) and Vp (posterior ventricular horn) measurement, demonstrating midline echo and cavum septum pellucidum (CSP)
Brain	Includes cerebellum with intact vermis, ventricles, cisterna magna (< 10mm) & cerebral hemispheres	Transcerebellar dimeter and nuchal fold
Face	Coronal view & profile. Normal nose and lips, chin in line with forehead, normal orbits.	Profile of the face Coronal view of lips with nasal tip
Spine	Confirm integrity throughout in transverse, sagittal & coronal sections. See each vertebra separately and	Sagittal and coronal views of the cervical, thoracic, lumbar and sacral spine with skin covering

	confirm normal curvature & skin covering.	
Neck/skin	No masses, nuchal fold < 6 mms	Image if nuchal fold appears 6.0mm or more. Measure the distance between the outer border of the occipital bone and the outer skin edge
Thorax	Normal size & echogenicity, no masses or free fluid	Sagittal view of chest to demonstrate heart, diaphragm and stomach
Heart	Left sided, fills approximately 1/3 of the fetal thorax. The right and left atrial chambers are a similar size. The right ventricular cavity is approximately equal to the left ventricular cavity across the inlet. The posterior walls of the right and left ventricles and the interventricular septum are approximately equal in thickness. The atrioventricular valves open and close with each cycle. The atrioventricular valves meet at the crux of the heart in an offset cross. The ventricular septum appears intact. The aorta and pulmonary artery are seen arising from the left ventricle and right ventricle respectively. They should cross over above the heart in a normal pattern, and should be of similar size at 19 wks (3-5mms). This can be seen in the 3 vessel view *In cases where there is a higher risk of congenital heart disease, the scan is recorded for review by the lead sonographer. In this case colour should be used for the 4 chamber & great artery views. See guideline “Referral for fetal echocardiogram”.	a) 4 chamber view b) Aorta (AO)/LVOT arising from the left ventricle c) Pulmonary artery (PA)/RVOT arising from the right ventricle/3 vessel view (3VV) d) 3 vessel and trachea view (3VT) e) situs of heart and stomach on split image and laterality of the heart with right and left labelled on the image
Abdomen	Free of masses, normal echogenicity. Abdominal wall intact.	At level of AC demonstrating stomach and short intra-hepatic section of the umbilical vein
GIT	Normal size, left sided stomach. No dilated bowel loops. The echogenicity of the bowel should be assessed for normality. The diaphragm should be seen intact separating the abdomen from the chest.	No images

Kidneys	Check size and echogenicity. If the renal pelves are prominent, they should be measured. If > 7mm see section 3.7 above.	Kidneys in transverse section bladder
Genitalia	Recorded only if of clinical relevance. Do not reveal sex unless the woman asks and only then if certain. Parents should be told that there is a small chance of error.	No images unless of clinical relevance
Extremities	Confirm 4 limbs. Fingers and toes do not need to be counted. Talipes and major hand anomalies should be excluded. All limb long bones should be seen to be of normal length, in proportion to each other and the fetal size.	Images to demonstrate femurs, tibia and fibula in both lower limbs, metatarsals in the feet, humeri, radius and ulna in forearms and metacarpals
Skeleton	Free from fractures, normal echogenicity	No specific images

Appendix 2

FASP 11 auditable conditions

Edwards' syndrome (T18)
Patau's syndrome (T13)
Anencephaly
Spina bifida
Cleft lip
Congenital diaphragmatic hernia
Gastroschisis
Exomphalos
Congenital heart disease
Bilateral renal agenesis
Lethal skeletal dysplasia